SECURITY ENGINEERING RESEARCH POSTER

A Unified Control Plane for Machine-Learning Security Services

An Authenticated Gateway for Independently Released ML Security Products

github.com/poojakira/unified-ml-security-platform - public, default branch main.

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

82	95.96%	4	Contract stubs
Core Python tests passed	Core statement coverage	Production synchronous service slots	Local product topology

SYSTEM VIEW

Architecture and Threat-Control Flow

Trust Boundary

Input surface
Data, models, requests, policies, or artifacts entering the tested path.

Control surface
Repository-specific validation, analysis, and fail-closed checks.

Evidence surface
Findings, metrics, tests, logs, or reproducible artifacts.

This diagram describes the tested repository path, not a universal deployment guarantee.

01 **Authenticate external callers at the gateway.** →

02 **Strip caller-controlled sensitive headers and inject per-service credentials.** ↓

03 **Route four synchronous service prefixes over an internal-only network.** →

04 **Use local health-contract stubs to test topology without misrepresenting product logic.** ↓

05 **Require explicit external product images in production Compose.** →

06 **Validate unit, security, integration, image-build, and deployment-contract behavior in CI.**

Research Contribution

A real authenticated FastAPI control plane with four production HTTP service slots, a local contract-test topology built from stubs, and a production Compose contract that requires externally released product images instead of pretending the stubs are product implementations.

Verified Evidence

EV-01	82 tests passed.	**EV-02**	95.96% statement coverage.
EV-03	Separate local contract-stub health jobs for MCP, LLM red-team, dataset-poison, and model-privacy completed successfully.	**EV-04**	Integration topology tests completed successfully.
EV-05	Security scan, dependency audit, production Compose validation, and gateway image build/push job completed successfully.	**EV-06**	Production product images: Externally supplied; stubs not published

Method and Control Path

01 **Authenticate external callers at the gateway.**

02 **Strip caller-controlled sensitive headers and inject per-service credentials.**

03 **Route four synchronous service prefixes over an internal-only network.**

04 **Use local health-contract stubs to test topology without misrepresenting product logic.**

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Limitations and Residual Risk

- Product business logic is owned by external repositories/images.
- No production deployment, uptime, or load SLO is established.
- Multi-replica shared rate limiting/state distribution requires deployment-specific infrastructure.
- Batch tools such as HF provenance scanning and adversarial robustness are intentionally not forced behind synchronous HTTP routes.

Reproducibility

```
Expected pinned CI Python 3.12 evidence: 82 passed,
95.96% statement coverage. The local Compose product
containers are contract stubs, not real downstream
products. Measured code snapshot:
d54336b5f5f058a4d756bf2c7a023ad8510bbd6f Evidence CI run:
37173681100
```

Evidence Provenance

Claim discipline
Metrics come from the repository's verified metrics and research brief.

Scope discipline
Benchmarks are reported only at their documented fixture, dataset, or test scope.

Reproduction
PDF and PNG are generated locally from committed evidence without paid services.

Evidence-backed security engineering

README, verified metrics, tests, CI, and committed artifacts remain authoritative. Poster layout emphasizes architecture, controls, evidence, limitations, and reproduction rather than unsupported efficacy claims.

36 x 48 in
PDF + PNG parity